DTC P0A94:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

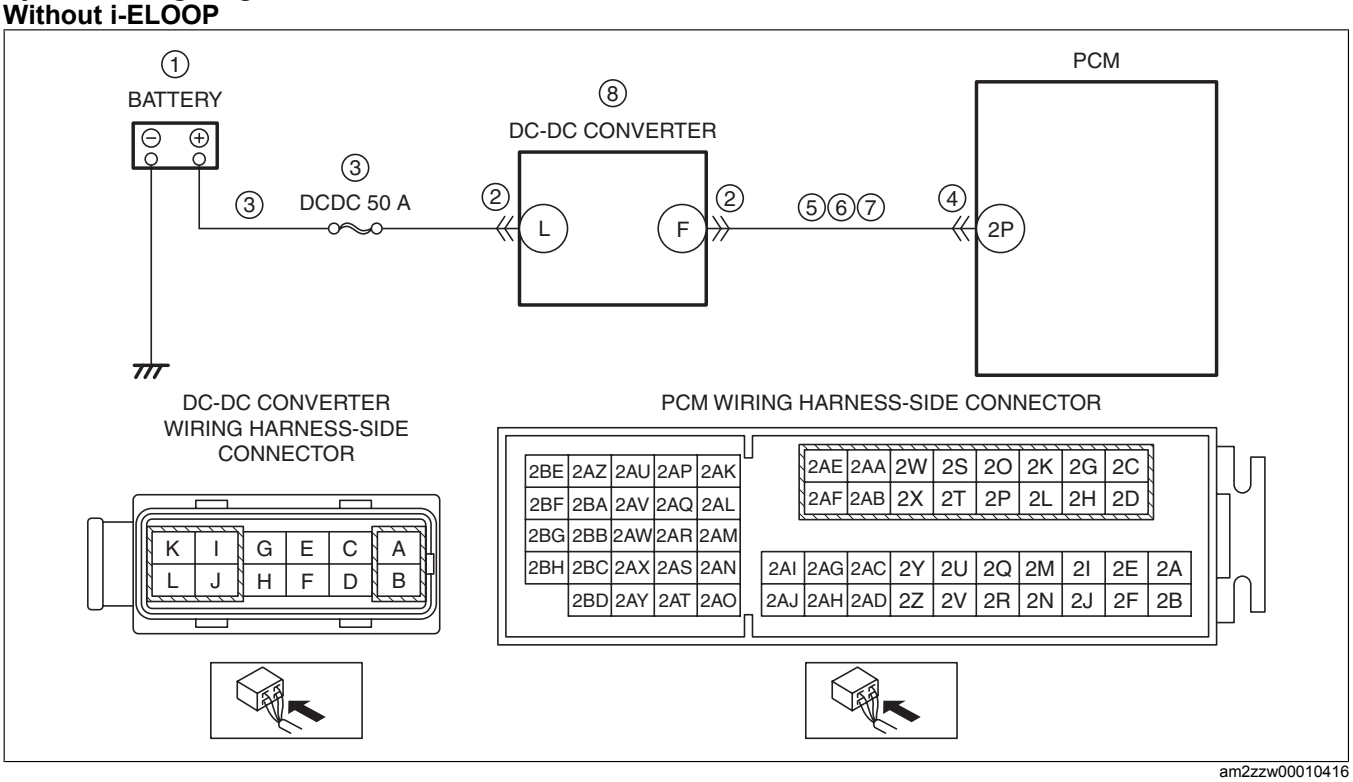
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Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".

Details On DTCs

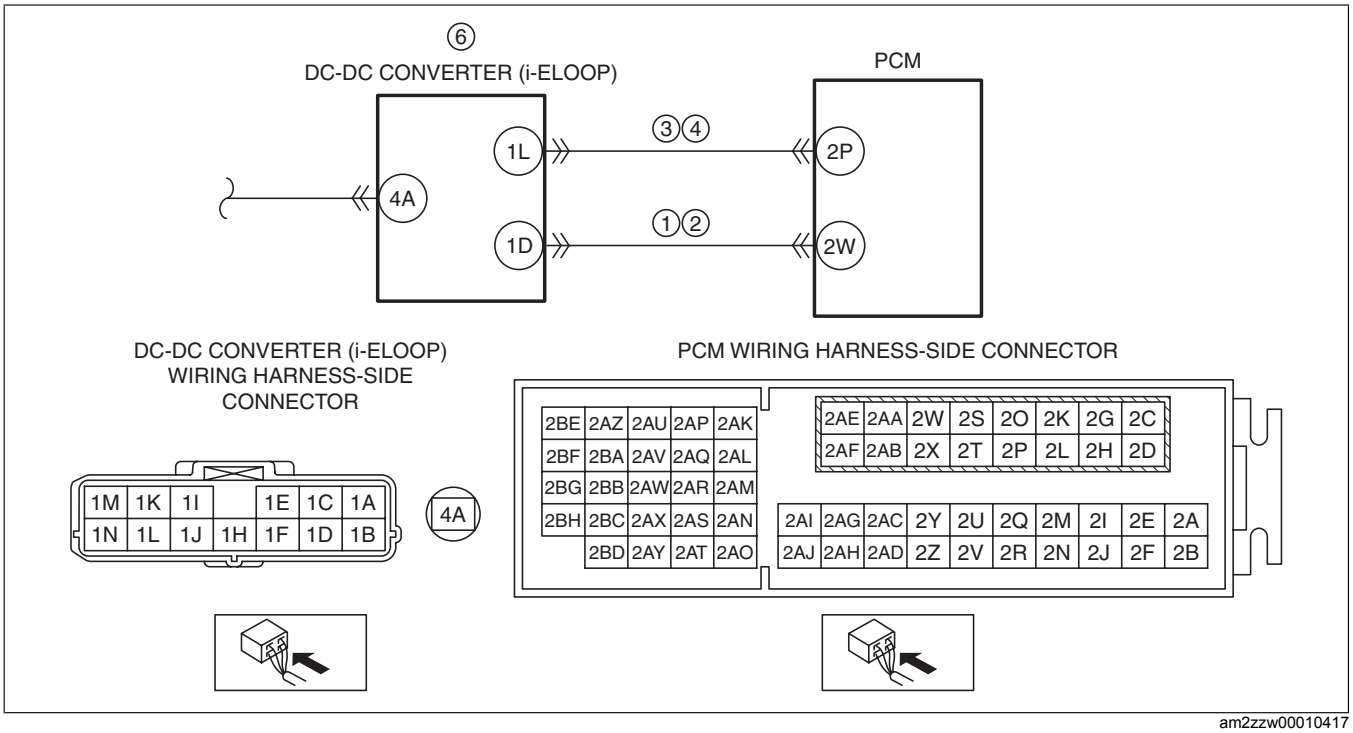
DESCRIPTION	**DC-DC converter: control circuit signal error (Without i-ELOOP)** **DC-DC converter (i-ELOOP): control circuit signal error (With i-ELOOP)**	
DETECTION CONDITION	Determination conditions	**Without i-ELOOP** • Internal malfunction signal from DC-DC converter via BCM is received. (CAN/LIN communication). • Input signal from the DC-DC converter limits the pressure increase time. • Input signal from the DC-DC converter does not implement pressure increase after a pressure increase command to the DC-DC converter. **With i-ELOOP** • The PCM receives a response signal from the DC-DC converter (i-ELOOP) which differs from the relay operation command voltage of the PCM.
	Preconditions	• Not applicable
	Drive cycle	• **1**
	Self test type	• CMDTC self test
	Sensor used	• Not applicable
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Depending on the type of malfunction, any of the following controls may be implemented. (With i-ELOOP) — Transition to i-ELOOP bypass mode. — i-ELOOP bypass mode inhibited.	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). (With i-stop) • The following vehicle conditions differ depending on the type of malfunction (With i-ELOOP) — Illuminates the master warning light — Illuminates the charging system warning light	
POSSIBLE CAUSE	**Without i-ELOOP** • DC-DC converter connector or terminals malfunction • Short to ground or open circuit in DC-DC converter power supply circuit — Short to ground in wiring harness between DCDC 50 A fuse and DC-DC converter terminal L — DCDC 50 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and DC-DC converter terminal L • PCM connector or terminals malfunction • Short to ground in wiring harness between DC-DC converter terminal F and PCM terminal 2P • Short to power supply in wiring harness between DC-DC converter terminal F and PCM terminal 2P • Open circuit in wiring harness between DC-DC converter terminal F and PCM terminal 2P • Battery malfunction • DC-DC converter malfunction • BCM malfunction • PCM malfunction **With i-ELOOP** • Connector or terminal malfunction of the following parts: — PCM — DC-DC converter (i-ELOOP) • Short to ground in wiring harness between the following terminals: — DC-DC converter (i-ELOOP) terminal 1L—PCM terminal 2P — DC-DC converter (i-ELOOP) terminal 1D—PCM terminal 2W • Open circuit in wiring harness between the following terminals: — DC-DC converter (i-ELOOP) terminal 1L—PCM terminal 2P — DC-DC converter (i-ELOOP) terminal 1D—PCM terminal 2W • DC-DC converter (i-ELOOP) malfunction • PCM malfunction	

System Wiring Diagram
Without i-ELOOP



am2zzw00010416

With i-ELOOP



am2zzw00010417

Function Explanation (DTC Detection Outline)
Without i-ELOOP

- The DC-DC converter is equipped with a boost function to stabilize the power supply to the vehicle's electrical devices when the engine is restarted by i-stop. If the DC-DC converter does not operate, the supply voltage to the vehicle's electrical devices decreases. In this diagnostic, the response condition of the DC-DC converter to the boost demand from the PCM, or the boost by the on-board diagnostic function of the DC-DC converter is not detected, and verification of vehicle malfunctions/safety assurance is performed.

With i-ELOOP

- The PCM performs malfunction determination of the relay ON/OFF command function for the DC-DC converter (i-ELOOP) for verification of vehicle malfunction/safety assurance.
  - The DC-DC converter (i-ELOOP) is equipped with a bypass circuit which supplies voltage generated by the generator to the **12 V** line. If this function does not operate, a power supply shortage occurs.
  - With this diagnosis, a malfunction is determined by the PCM, in which a bypass circuit of the DC-DC converter (i-ELOOP) from the PCM is on or the response signal of the DC-DC converter (i-ELOOP) related to the off command voltage does not match.
  - To stabilize the power supply of the electrical devices when the engine is restarted by i-stop, the DC-DC converter (i-ELOOP) is equipped with a function which disconnects the power supply of the electrical devices from the battery and supplies electricity directly through the DC-DC converter (i-ELOOP). If this function does not operate, the power supply of the electrical devices is turned off instantaneously due to the effect of cranking when the engine is restarted by i-stop. (with i-stop)
  - With this diagnosis, a malfunction is determined by the PCM, in which an i-stop relay circuit of the DC-DC converter (i-ELOOP) from the PCM is on or the i-stop relay circuit operation condition of the DC-DC converter (i-ELOOP) related to the off command voltage does not match. (with i-stop)

Repeatability Verification Procedure

Without i-ELOOP

1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Start the engine.
3. Stop the engine by operating the i-stop function.

With i-ELOOP

1. Disconnect the negative battery cable and wait for **10 s or more**.
2. Connect the negative battery cable, start the engine, and maintain the idle status.

PID Item/Simulation Item Used In Diagnosis

- Not applicable

Function Inspection Using M-MDS (Without i-ELOOP)

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.
3	**PURPOSE: VERIFY IF BATTERY VOLTAGE IS FALSELY RECOGNIZED BY DTC RELATED CAN OR LIN COMMUNICATION** • Switch the ignition off, then ON (engine off). • Perform the PCM and BCM DTC inspection using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC INSPECTION [BCM].) • Are DTCs related CAN or LIN communication recorded?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [BCM].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
4	**PURPOSE: VERIFY IF THE CURRENT SENSOR IS MISIDENTIFYING MALFUNCTIONS DUE TO A BCM MALFUNCTION** • Perform the BCM DTC inspection using the M-MDS. (See DTC INSPECTION [BCM].) • Are any DTCs present?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure (Without i-ELOOP)
Intention of troubleshooting procedure

- Step 1
  - — Perform a unit inspection of the battery.
- Step 2—7
  - — Verify if a malfunction is occurring due to a malfunction in each signal transmission and the boost demand line.
- Step 8—9
  - — Verify that primary malfunction is resolved and there are no other malfunctions.

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**PURPOSE: INSPECT BATTERY** • Switch the ignition off. • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Recharge or replace the battery, then go to Step 8. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
2	**PURPOSE: INSPECT DC-DC CONVERTER CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the DC-DC converter connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**PURPOSE: INSPECT DC-DC CONVERTER POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the DC-DC converter connector is disconnected. • Measure the voltage at the DC-DC converter terminal L (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the DCDC 50 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between DCDC 50 A fuse and DC-DC converter terminal L. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and DC-DC converter terminal L. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.
4	**PURPOSE: INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
5	**PURPOSE: INSPECT DC-DC CONVERTER SIGNAL CIRCUIT FOR SHORT TO GROUND** • Verify that the DC-DC converter and PCM connectors are disconnected. • Inspect for continuity between DC-DC converter terminal F (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter terminal F and PCM terminal 2P. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 8.
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**PURPOSE: INSPECT DC-DC CONVERTER SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the DC-DC converter and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the DC-DC converter terminal F (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter terminal F and PCM terminal 2P. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 8.
7	**PURPOSE: INSPECT DC-DC CONVERTER SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Verify that the DC-DC converter and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between DC-DC converter terminal F (wiring harness-side) and PCM terminal 2P (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter terminal F and PCM terminal 2P. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
8	**PURPOSE: VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Replace the DC-DC converter, and then perform Step 8 again. (See DC-DC CONVERTER REMOVAL/INSTALLATION [WITHOUT i-ELOOP].) • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
9	**PURPOSE: VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.

Function Inspection Using M-MDS (With i-ELOOP)

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
2	**PURPOSE: RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure (With i-ELOOP)
Intention of troubleshooting procedure

- Step 1—2
  - — Verifying malfunction related to DC-DC converter (i-ELOOP) 1D terminal
- Step 3—4
  - — Verifying malfunction related to DC-DC converter (i-ELOOP) 1L terminal
- Step 5—6
  - — Verifying/replacing malfunctioning part
- Step 7—8
  - — Verify that primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY IF SHORT TO GROUND IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Disconnect the connector of the following parts. — PCM — DC-DC converter (i-ELOOP) • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — DC-DC converter (i-ELOOP) terminal 1D • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • PCM terminal 2W—DC-DC converter (i-ELOOP) terminal 1D **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 7.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF OPEN CIRCUIT IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the DC-DC converter (i-ELOOP) and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — PCM terminal 2W—DC-DC converter (i-ELOOP) terminal 1D • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • PCM terminal 2W—DC-DC converter (i-ELOOP) terminal 1D **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.

STEP	INSPECTION	RESULTS	ACTION
3	**PURPOSE: VERIFY IF SHORT TO GROUND IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the DC-DC converter (i-ELOOP) and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — DC-DC converter (i-ELOOP) terminal 1L • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • PCM terminal 2P—DC-DC converter (i-ELOOP) terminal 1L **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 7.
		No	Go to the next step.
4	**PURPOSE: VERIFY IF OPEN CIRCUIT IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the DC-DC converter (i-ELOOP) and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — PCM terminal 2P—DC-DC converter (i-ELOOP) terminal 1L • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • PCM terminal 2P—DC-DC converter (i-ELOOP) terminal 1L **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.
5	**PURPOSE: VERIFY IF MALFUNCTION OCCURRING IS CAUSED BY WIRING HARNESS/ CONNECTOR LOSS** • Have all the wiring harnesses/connectors which can be considered the cause of DTC P0A94:00 been inspected?	Yes	Go to the next step.
		No	Inspect all the related wiring harnesses/connectors. **If there is a malfunction** • Repair or replace the connector and/or terminals, then go to Step 7. **If there is no malfunction** • Go to the next step.
6	**PURPOSE: VERIFY DC-DC CONVERTER (i-ELOOP) REPLACEMENT RECORD** • Is the DC-DC converter (i-ELOOP) being replaced within the troubleshooting diagnostic procedure this time?	Yes	Replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Replace the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) Go to the next step.
7	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 5.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
8	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.